

1. 10 different model providers and model released in last 12 months

Provider	Model	Release Date	Cutoff Date	Cost of Tokens	Context Window	Type	When to Use
OpenAI	GPT-5.6 (Sol, Terra, Luna)	Jun 26, 2026	Feb-26	\$5 / \$25 per Mtok	1M	Frontier multimodal	Best for reasoning, coding, tool use, enterprise workloads
Anthropic	Claude Opus 5	Jul 24, 2026	Feb-26	\$5 / \$25 per Mtok	1M	Frontier multimodal	Knowledge work, automation, long-horizon reasoning
Google DeepMind	Gemini 3.6 Flash	Jul 21, 2026	Feb-26	~\$0.03 / \$0.13 per Mtok	1M	Multimodal	Fast multimodal tasks, cost-efficient agents
Mistral AI	Mistral Medium 3.5	Apr 29, 2026	Jan-26	\$0.50 / \$1.50 per Mtok	256K	Open-weight multimodal	Open-source coding, reasoning, multilingual
Meta AI	Muse Spark 1.1	Jul 9, 2026	Feb-26	Proprietary API pricing	1M	Multimodal reasoning	Agentic tool use, multimodal reasoning
Cohere	Command A+	May 20, 2026	Jan-26	Apache 2.0, cheap inference	128K	MoE multimodal	Enterprise-grade multilingual, agentic tasks
xAI (SpaceXAI)	Grok 4.5	Jul 8, 2026	Feb-26	\$2 / \$6 per Mtok	500K	Frontier coding + chat	Coding, agentic workflows, reasoning
AI21 Labs	Jamba2	Jan 8, 2026	Nov-25	Open-weight	256K	Hybrid SSM-Transformer	Long-context reasoning, enterprise text tasks

Alibaba (Qwen)	Qwen3.7-Flash	Jul 27, 2026	Feb-26	\$0.03 / \$0.13 per Mtok	1M	MoE multimodal	Cost-efficient multimodal agents
DeepSeek	DeepSeek V4-Flash	Jul 31, 2026	Feb-26	\$0.14 / \$0.28 per Mtok	1M	MoE open-weight	Coding, reasoning, tool use at low cost

Key Points:

OpenAI GPT-5.6 and **Claude Opus 5** are the strongest frontier models, ideal for enterprise reasoning and automation.

Google Gemini 3.6 Flash and **Qwen3.7-Flash** stand out for **cost efficiency** in multimodal workloads.

Mistral Medium 3.5 and **DeepSeek V4-Flash** are **open-weight models**, great for developers who want transparency and local deployment.

Meta Muse Spark 1.1 is Meta's first closed Muse model, focusing on multimodal reasoning and agent orchestration.

Cohere Command A+ emphasizes **multilingual and agentic tasks**, with Apache 2.0 licensing making it flexible for enterprise use.

xAI Grok 4.5 is tuned for **coding and agentic workflows**, with configurable reasoning effort.

AI21 Jamba2 continues their hybrid SSM-Transformer line, optimized for **long-context reasoning**.

Risk:

Closed vs Open Weights: OpenAI, Anthropic, Meta, and xAI mostly keep models proprietary, limiting transparency. Mistral, Cohere, DeepSeek, and Alibaba offer open-weight options.

Cost Variability: Token pricing differs widely — OpenAI and Anthropic are premium, while Alibaba and DeepSeek undercut with aggressive pricing.

Context Windows: Models now reach **1M tokens** (OpenAI, Anthropic, Google, Alibaba, DeepSeek), but practical performance varies.

Specialization: Some models (e.g., Cohere Command A+, Grok 4.5) are tuned for coding/agentic tasks, while others (Gemini, Muse Spark) excel at multimodal reasoning.

2. 2 to 3 Articles about transformer works

- (a) Self-attention mechanism
- (b) Vector embedding
- (c) Feed forward Neural Network

Here the Article/blog referred for Transformers

[The Illustrated Transformer – Jay Alammar – Visualizing machine learning one concept at a time.](#)

[Transformers v5: Simple model definitions powering the AI ecosystem](#)

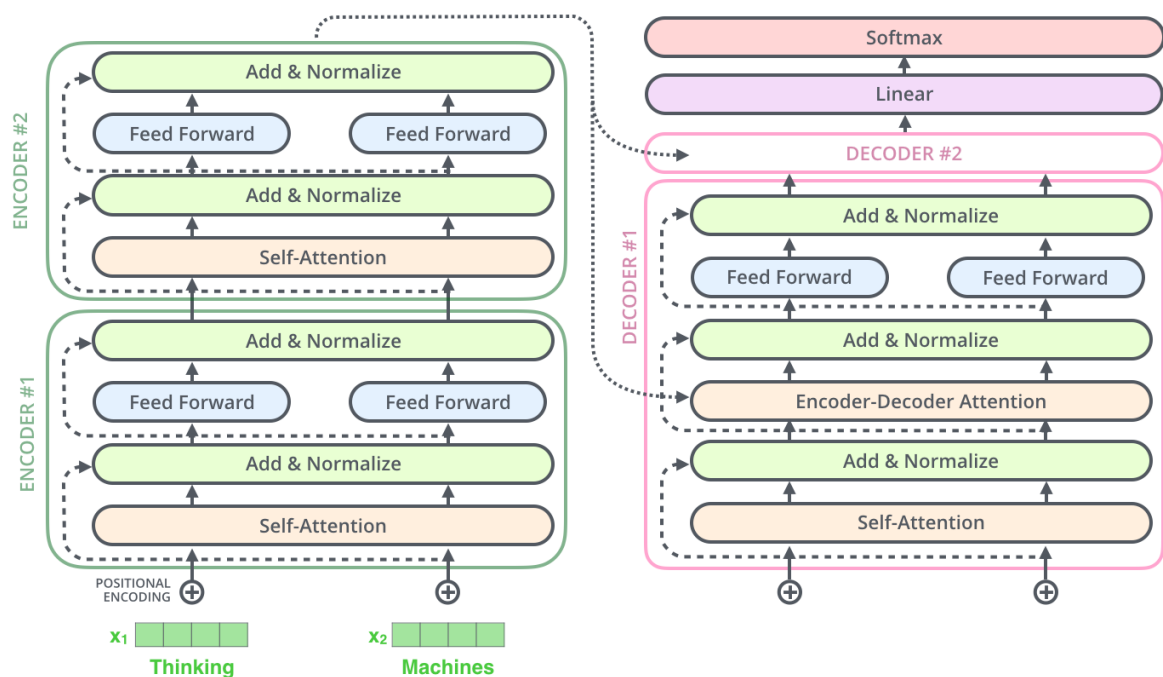
<https://towardsai.net/p/nlp/transformers-explained-visually-part-1>

Jay Alammar's Illustrated Transformer for the **visual intuition**.

Hugging Face blog for a **practical, code-oriented perspective**.

Towards AI article for a **structured, visual explanation**.

Here sample architecture diagram from which I referred.



1. Vector Embedding

Convert input tokens into numerical vectors that capture meaning.

- Tokenize the input text into words or sub words
- Map each token to a dense vector using an embedding matrix
- These vectors represent semantic meaning in high-dimensional space

2. Self-Attention Mechanism

Core Process - Each token attends to all others to capture relationships.

- Compute Query, Key, Value vectors from embeddings

- Calculate attention scores: dot product of Query and Key
- Apply softmax to get attention weights
- Weighted sum of Values produces context-aware representation

3. Feed Forward Neural Network

Refinement - Transform attended representations with non-linear layers.

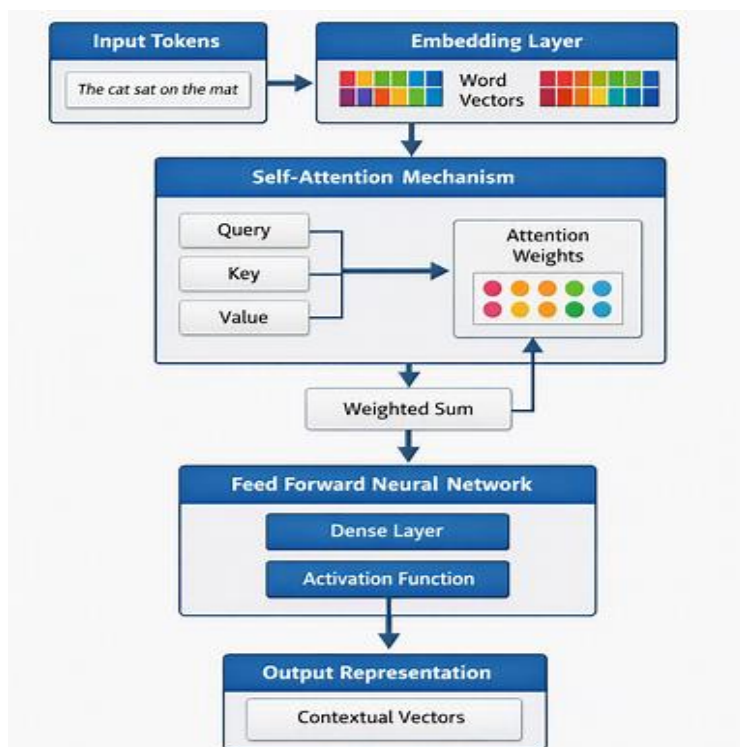
- Pass each token's representation through dense layers
- Apply activation functions (e.g., ReLU or GELU)
- Add residual connections and layer normalization
- Produces richer, task-ready representations

4. Output Layer

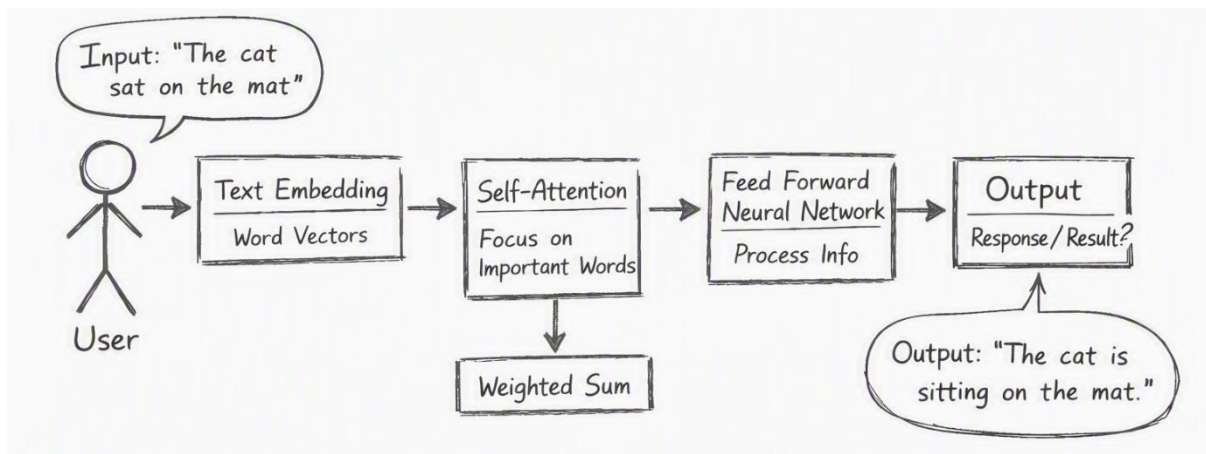
Final Step - Generate predictions or pass to next transformer block.

- For language tasks: predict next token probabilities
- For classification: output class scores
- For translation/summarization: produce sequence outputs

Transformer flow diagram:



Example:



Key points:

Input Tokens → text broken into tokens.

Embedding Layer → converts tokens into numerical vectors.

Self-Attention Mechanism → computes Query, Key, and Value vectors, then attention weights.

Feed Forward Neural Network → applies dense layers and activation functions.

Output Representation → contextual vectors ready for prediction or next layer